

May 5

Ex. of applying thms. 14.3, 14.4 Prove $\langle x \rangle \in \mathbb{Z}[x]$ is a prime ideal that is not maximal

Consider $\mathbb{Z}[x]/\langle x \rangle$. "When quotienting, everything in the ideal becomes 0" $\rightarrow \mathbb{Z}[x]/\langle x \rangle = \mathbb{Z}[0] \cong \mathbb{Z}$

Let $a_n x^n + \dots + a_1 x + a_0 + \langle x \rangle$ be an arbitrary coset. We see $a_n x^n + \dots + a_1 x \in \langle x \rangle$, so $a_n x^n + \dots + a_1 x + a_0 + \langle x \rangle = a_0 + \langle x \rangle$.

Thus every coset has the form $a + \langle x \rangle$ for $a \in \mathbb{Z}$. If $a \neq b$, then $a + \langle x \rangle \neq b + \langle x \rangle$ since $a - b \notin \langle x \rangle$. $aN = bN \Leftrightarrow a - b \in N$

Thus $\mathbb{Z}[x]/\langle x \rangle$ is pretty much the same as $\mathbb{Z}$. Since $\mathbb{Z}$ is an integral domain, we see $\langle x \rangle$ is prime (thm. 14.3).
isomorphic, but we haven't defined that for rings

Since $\mathbb{Z}$ is not a field, we see $\langle x \rangle$ is not maximal (thm. 14.4).

• Thm. 14.3 Let R be a commutative ring with 1, and let A be an ideal of R . Then R/A is an integral domain $\Leftrightarrow A$ is prime

Proof

($\Leftarrow$) Suppose A is prime. Let $x+A, y+A$ be arbitrary elements of R/A . In other words, $x, y \in R$ are arbitrary. Suppose $(x+A)(y+A) = 0+A$. By def. $(x+A)(y+A) = (xy)+A$. Thus $xy+A = 0+A$. This implies $xy \in A$. Since A is prime, $x \in A$ or $y \in A$. Thus $x+A = 0+A$ or $y+A = 0+A$. So R/A is an integral domain.

($\Rightarrow$) Suppose R/A is an integral domain. Let $x, y \in R$ such that $xy \in A$. Thus $xy+A = 0+A$. Hence $(x+A)(y+A) = 0+A$. Since R/A is an integral domain, $x+A = 0+A$ or $y+A = 0+A$. Thus $x \in A$ or $y \in A$. So, A is prime. $\square$

• Thm. 14.4 Let R be a commutative ring with 1, and let A be an ideal of R . Then R/A is a field $\Leftrightarrow A$ is maximal

Proof

($\Rightarrow$) Suppose R/A is a field and suppose B is an ideal of R such that $A \subseteq B$. Want $B = R$.

Let $b \in B - A = B \setminus A$, set minus. Thus $b+A$ is a non-identity element of R/A . Since R/A is a field, $\exists c+A \in R/A$ such that $(b+A)(c+A) = 1+A$. Thus $bc+A = 1+A$. Hence $bc + a = 1$ for some $a \in A$. Now B is an ideal containing A . Thus $a \in B$ and $bc \in B$, so $bc + a \in B$. i.e. $1 \in B$. This implies $B = R$. Indeed, if $r \in R$, then $r \cdot 1 = r \in B$, so $R \subseteq B$ and $R = B$. We see that A is maximal.

($\Leftarrow$) Suppose A is maximal. Let $b \in R - A$. We wish to show $b+A$ has a multiplicative inverse in R/A . Consider the set $B = \{br + a : r \in R, a \in A\}$. We claim* this is an ideal of R that properly contains A . Since A is maximal, $B = R$. In particular, $1 \in B$. Thus $1 = br + a$ for some $r \in R$ and some $a \in A$. Thus $1+A = br+A = (b+A)(r+A)$. Hence $b+A$ has a multiplicative inverse in R/A , so R/A is a field. $\square$

* Ex. Prove $B = \{br + a : r \in R, a \in A\}$ is an ideal of R properly containing A .

We apply the ideal test.

Nonempty

$0 = b(0) + 0 \in B$ since R and A are rings, and therefore must contain 0 . So B is nonempty.

Closure under subtraction

Let $b_1, b_2 \in B$. We see $b_1 = br_1 + a_1$, $b_2 = br_2 + a_2$ for some $r_1, r_2 \in R$ and $a_1, a_2 \in A$. We compute $b_1 - b_2 = (br_1 + a_1) - (br_2 + a_2) = br_1 + a_1 - br_2 - a_2 = b(r_1 - r_2) + (a_1 - a_2) \in B$. So B is closed under subtraction.

Multiplicative absorption

Let b_1 be as above and $r \in R$. Then $b_1 \cdot r = (br_1 + a_1)r = br_1r + ar \in B$. So, B satisfies multiplicative absorption.

We conclude B is an ideal.

To show $A \subsetneq B$, observe that $\forall a \in A$, $a = b \cdot 0 + a \in B$. So $A \subset B$. However, $b \cdot 1 + 0 = b \notin A$ by construction ($b \in B - A$), so $B \neq A$.

So, the claim holds. $\square$

Ch. 15 Ring Homomorphisms and the First Isomorphism Theorem

• Def A ring homomorphism is a function $\phi: R \rightarrow S$ between rings R and S such that

- 1) $\phi(a+b) = \phi(a) + \phi(b)$
 - 2) $\phi(ab) = \phi(a) \cdot \phi(b)$
- $\} \forall a, b \in R, \text{ "}\phi \text{ respects the ring operations"}$

• If, in addition, ϕ is bijective, then we say ϕ is a ring isomorphism. In this case, the structure of S is the same as that of R and we write $R \cong S$.

Ex. $\mathbb{Z}[i]/\langle i-1 \rangle \cong \mathbb{Z}/2\mathbb{Z}$ from HW

$$\mathbb{R}[x]/\langle x^2+1 \rangle \cong \mathbb{C}$$

• Examples of homomorphisms:

• Reduction: $\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z}$ given by $a \mapsto \bar{a} = a + m\mathbb{Z} = a + \langle m \rangle$

• $\iota: \mathbb{Z} \hookrightarrow \mathbb{Q}$ given by $a \mapsto a = [a, 1]$ under rational equivalence
 $\hookrightarrow$ canonical embedding of $\mathbb{Z}$ into $\mathbb{Q}$

• $\text{Eval}_1: \mathbb{Z}[x] \rightarrow \mathbb{Z}$ given by $f(x) \mapsto f(1)$, $x \mapsto 1$, i.e. $x^3 - 1 \mapsto 1^3 - 1 = 0$

$$\begin{aligned} \text{Ker Eval}_1 &= \{f(x) \in \mathbb{Z}[x] : f(1) = 0\} \\ &= \{f(x) \in \mathbb{Z}[x] : (x-1) \mid f(x)\} \\ &\quad \downarrow \\ &\quad 0 \text{ in } S \end{aligned}$$

• Def If $\phi: R \rightarrow S$ is a ring homomorphism, the kernel of ϕ is the set $\text{Ker } \phi = \{r \in R : \phi(r) = 0_S\}$

• Def The image of a homomorphism is $\phi(R) = \{s \in S : \phi(r) = s \text{ for some } r \in R\}$

• Thms. 15.2 & 15.4 $\text{Ker } \phi$ is an ideal and every ideal looks like $\text{Ker } \phi$ for some homomorphism ϕ

• Thm. 15.3 (The first isomorphism theorem for Rings) If $\phi: R \rightarrow S$ is a ring homomorphism, then $R/\text{Ker } \phi \cong \phi(S)$

Ex. Define $\phi: \mathbb{Z}[x] \rightarrow \mathbb{Z}[i]$ by $\phi(x) = i$. We can see ϕ is surjective. ($a+bx \mapsto a+bi$)
 $\text{Ker } \phi = \{f(x) \in \mathbb{Z}[x] : f(i) = 0\}$. With some work, we can show $\text{Ker } \phi = \langle x^2 + 1 \rangle$. By the FIT, $\mathbb{Z}[x]/\langle x^2 + 1 \rangle \cong \mathbb{Z}[i]$

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